

MODELING CHOICE INTERDEPENDENCE IN A SOCIAL NETWORK

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ABSTRACT

This paper shows how researchers can model choice interdependence in a social network where each individual's choice decision can be influenced by the decisions of connected others. To properly account for the dependence structure in the data, we propose a Markov Random Field (MRF) to model choice interdependence when only choices are observed but the choices' sequence is not and a discrete-time Markov chain when both the choices and their sequence are observed. We compare two estimation methods, pseudolikelihood and approximate sampling, to circumvent the intractable normalizing constant problem in estimating the MRF model. We also show the theoretical connection between the MRF and the discrete-time Markov chain. The findings from our field experiment show that consumers rely on choices of other both at the micro (i.e., closely connected friends) and macro (i.e., a reference group) levels in making their own decisions. Both of the proposed models can also accommodate asymmetry in choice interdependence between pairs of individuals.

Keywords: choice interdependence, social network, Markov random field, Bayesian

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Although many situations call for consumers to make independent individual decisions, realistically, every individual decision is likely influenced by the decisions of others. For example, individual decisions to purchase a hybrid vehicle might be interdependent with car choice decisions made by others in their social circles. An individual's decision to join a fan page on Facebook may also depend largely on whether his or her Facebook friends have already joined. Thus, a challenging question is how to measure two distinct sources of underlying preference driving such choice interdependence in a social network: inherent individual preference and choices made by their connected others.

We primarily focus on modeling choice interdependence as a reflection of changes in consumers' preferences induced by the choices made by others in a social network. Our paper deviates from the related literature in several ways. First, unlike the prior literature that attempts to model the impact of online word-of-mouth (Chevalier and Mayzlin 2006; Godes and Mayzlin 2004; Zhu and Zhang 2010), viral marketing activities (Leskovec, Adamic, and Huberman 2007; Toubia, Stephen and Freud 2009), and online community participation (Algesheimer et al. 2010; Stephen and Toubia 2010) on aggregate sales, our interest lies in modeling choice interdependence at the individual level. Second, our focus is on capturing how an individual consumer's choice is influenced by others' choices in a social network, and not on deriving group choices from diverse individual preferences (Aribarg, Arora and Bodur 2002; Aribarg, Arora and Kang 2010) or on modeling coordinated choices (Hartmann 2009; Yang et al. 2009) among a small group of consumers. Third, our source of interdependence is *choices* made by others, not *preferences* of others (Yang and Allenby 2003; Yang, Narayan and Asseal 2006) in a network. Finally, similar to Nair, Manchanda and Bhatia (2009), our models can accommodate asymmetric effects of social influence. However, our approach aims at modeling the joint distribution of individual choices within a network, and not the marginal distribution (i.e., an ego-centric approach) of individual quantity decisions.

Modeling choice interdependence in a social network is a challenging problem. To properly account for the dependence structure in the data (i.e., the joint probability of individuals' choices in a network), we propose two modeling frameworks for two types of choice data researchers may collect. First, researchers may observe a cross-sectional snap-shot of realized choices in a network at a particular time point (hereafter referred to as snap-shot data). In this case, although actual choices are made sequentially by different people, the choice sequence cannot be observed (e.g., researchers can only observe choices of hybrid vehicles in a neighborhood at a particular point in time). We propose a Markov random field (MRF) model which can also accommodate multiple relations within a network. Yet a big hurdle of fitting an MRF model is its well-known intractable normalizing constant problem, that is, the normalizing constant is a function of parameter values and thus cannot simply be ignored in the derivation of the posterior density. We demonstrate how either a pseudolikelihood (Besag 1974, 1975) or approximate sampling (Wang and Atchadé 2010) approach can be used to simplify the estimation of an MRF model.

With the social media technology, it is also feasible to collect longitudinal choice data where choices made in a network are recorded real time, and thus researchers observe the sequence of choices (hereafter referred to as complete data). While snap-shot data can be a realization of multiple alternative dynamic processes, complete data carry the exact information about the dynamic process within the network that gives rise to the realized choices. Given its complete information on the choice evolution, we show that researchers need a less complicated model, which does not suffer from the intractable normalizing constant problem, to fit complete data. We propose a discrete-time Markov chain model for this type of data. As snap-shot data are a result of a dynamic process, we also show sufficient conditions for the discrete-time Markov chain to converge to the MRF. This result suggests that MRF is a natural model that emerges from the Markov chain dynamic choice process.

To test our proposed model, we design a field experiment where we first collect

information about participants' social connections within a specific social network, as well as measuring each participant's intrinsic preferences. We later observe in real time how they make product choices given they observe other participants' choices. We fit the MRF model to the snap-shot data (i.e., realization of the complete data) and the discrete-time Markov chain model to the complete data. Controlling for individuals' heterogeneous preferences, our results show that participants rely on choices observed both at the micro (i.e., closely-connected friends) and macro (i.e., a reference group) levels in making their decisions. We also find asymmetry in choice interdependence across different types of consumers (i.e., expert vs. novice).

The paper is structured as follows. We start with presenting our modeling framework and theory. This is followed by a detailed description of our experimental design and data collection procedure. The estimation results and model comparison are included in the empirical analysis section. We close with a discussion about future research.

MODELING FRAMEWORK AND THEORY

To examine choice interdependence in a social network, two types of choice data can be collected: snap-shot and complete data. Snap-shot data are cross-sectional choice data collected at a fixed time point. This type of data is easier and less expensive to obtain. However, lack of information on the sequence of choices makes it difficult to understand the dynamic process, as different processes can give rise to the same snap-shot data. Complete data, on the other hand, are longitudinal choice data where the sequence of choices is also observed. With this additional information, we can better understand the role of a social network in the entire dynamic choice process, that is, not only can we model 'which' choice is made, but we can also model 'who' makes the choice. No matter what type of data is available, we need to model all consumers' choices jointly rather than independently to properly preserve the dependence structure in the

data.

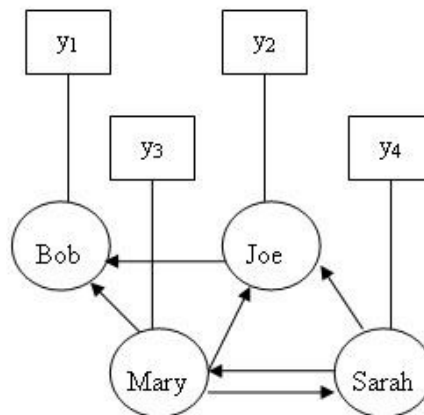
Markov Random Field (MRF) with Multiple Relations for Snap-shot Data

For snap-shot data, we can specify a Markov random field (MRF) to capture the interdependence among consumers' choices. Consumers' choices $y = (y_1, \dots, y_n)$, with y_i taking nominal values from $0, \dots, K$, are the variables constituting a random field, and their dependence comes from the presence of relationships among consumers in a social network (Figure 1). The circles and arrows in Figure 1 depict a directed network of consumers, where relationships are not necessarily reciprocal.

An MRF is a random field that satisfies the Markovian property, that is, the conditional probability of y_i only depends on i 's neighbors. Mathematically, it means

$p(y_i | y_{-i}) = p(y_i | y_j, j \in N(i))$, where $y_{-i} = (y_1, \dots, y_{i-1}, y_{i+1}, \dots, y_n)$, and $N(i)$ contains all i 's neighbors. These neighbors include anyone whose choice y_i may be conditionally dependent due to their social relationships, and they form a clique.

Figure 1
A Random Field of Consumers' Choices within a Social Network



Hammersley and Clifford's theorem (Hammersley and Clifford 1971) simplifies the specification of the MRF by proving that, conditional on the size of the clique, its joint distribution can be determined only by the interactions within it. Specifically, the joint distribution of y is given by:

$$(1) \quad p(y_1, \dots, y_n | \theta) = \frac{1}{Z(\theta)} \exp \left\{ \sum_{c \in C} U_c(y_i, i \in c; \theta) \right\},$$

where C is a set of relevant cliques, U_c is a function of model parameters technically called the potential of the clique (the parameters in U_c may vary from clique to clique), θ denotes all the

unknown parameters, and $Z(\theta) = \sum_{y_1, \dots, y_n} \exp \left\{ \sum_{c \in C} U_c(y_i, i \in c; \theta) \right\}$ is the normalizing constant, which is a summation over all possible values of vector y .

Our proposed MRF model can also accommodate multiple relations. The choice interdependence in real life is usually complicated and may result from different types of relationships. Therefore for each relationship, we specify a separate set of cliques for each individual. We conceptualize that each individual is influenced by both micro (i.e., closely-connected friends) and macro (e.g., a reference group) levels of relationships.

We specify the MRF with multiple relations as follows:

$$(2) \quad p(y_1, \dots, y_n | \theta) = \frac{1}{Z(\theta)} \exp \left\{ \sum_{m=1}^M \sum_{c \in C_m} U_c(y_i, i \in c; \theta) \right\},$$

where M is the number of relationships and C_m are possible cliques corresponding to relationship m . It is easy to see that $p(y_i | y_{-i}, \theta) = p(y_i | y_j, j \in \mathcal{N}_{m=1}^M N_m(i), \theta)$ with $N_m(i)$ i 's neighbors in m , which means the conditional distribution of y_i only depends on i 's neighbors.

Some remarks related to the MRF are worth mentioning. First, an MRF can accommodate networks in various formats. A directed network allows relationships to be non-

reciprocal. A directed and/or weighted network allows relationships to be asymmetric. Second, estimating an MRF does not require the inclusion of all possible cliques. However, in order to capture choice interdependence, some cliques of size greater than one need to be included, as cliques of size one are only related to individual unique characteristics.

Third, U_c can be any arbitrary function. An example of U_c in Figure 1 between Sarah and Joe could be $U_c = \theta I(y_2 = y_4)$, where I is the indicator function. Then a positive coefficient reflects the tendency to purchase the same product, if a consumer shares the purchase decision with another consumer. We can use the same potential for cliques of the same size. On the other hand, we can use different potentials for cliques (i, j) and (j, i) if the network is directed.

Fourth, with some necessary conditions, we can identify an asymmetric dependence structure with the MRF. The first condition involves a clearly defined identity of each node: at the individual level or at the role level. The second condition is that statistics corresponding to the parameters for asymmetry must have different values. To identify asymmetric dependence in the case of individuals, researchers also need to observe more than one choice for each individual. If nodes' identities are defined at the role level, one decision per individual will be sufficient as long as aggregate statistics associated with each role differ.

Finally, despite the ease and flexibility of modeling choices jointly, an MRF model has an intractable normalizing constant $Z(\theta)$. In the binary choice case, $Z(\theta)$ involves the summation of 2^n terms. The brute force calculation is only feasible for a very small n , which is often not of practical interest. Such a combinatoric problem is further exacerbated in the case of multinomial choice. In a Bayesian framework, draws from the posterior distribution for most models can be obtained easily, as its normalizing constant can be ignored, $p(\theta | y) \propto \pi(\theta)p(y | \theta)$, or gets cancelled out in the Hasting ratio. In the case of MRF, the normalizing constant (in equations 1 and 2) involves unknown parameters and hence cannot be ignored or cancelled out. As a result, the evaluation of the normalizing constant remains a challenging problem.

Estimation of the Markov Random Field

We introduce two estimation approaches to resolve the intractable normalizing constant problem: the pseudolikelihood approach and the approximate sampling approach. The pseudolikelihood approach replaces the likelihood with a pseudolikelihood function that does not involve an intractable normalizing constant, thus is easy to compute. However, the pseudolikelihood function is not the exact likelihood function, $p(y | \theta)$, hence the corresponding posterior distribution, $p(\theta | y)$, is also not the one of interest. In contrast, the approximate sampling approach works directly with the likelihood function and estimates the posterior distribution by approximating the normalizing constant in the Markov chain Monte Carlo (MCMC) procedure.

Pseudolikelihood Approach

The maximum pseudolikelihood estimator is proposed by Besag (Besag 1974, 1975) as an alternative to the maximum likelihood estimator. The pseudolikelihood function is defined as:

$$(3) \quad pl(y_1, \dots, y_n | \theta) = \prod_{i=1}^n p(y_i | y_{-i}, \theta).$$

Each term on the right is the conditional probability of y_i that only depends on i 's neighbors. The intractable normalizing constant problem does not occur in this specification because of this

conditional independent structure. In case of choice variable, $p(y_i | y_{-i}, \theta)$ has the same form as

a logit model except that y_{-i} enters as a covariate. To further illustrate the connection between the likelihood of an MRF and its corresponding pseudolikelihood, suppose we specify an MRF with three individuals facing four product options, where individuals 1 declares a relationship with 2 but not vice versa, and individuals 2 and 3 have reciprocal relationships. One possible specification based only on cliques of size one (i.e., capturing the utility each individual i can obtain for each choice option) and size two (e.g., capturing choice interdependence) is

$$p(y | \theta) = \frac{\exp\left\{\sum_{1 \leq i \leq 3} \beta u(y_i) + \theta[I(y_1 = y_2) + 2 \times I(y_2 = y_3)]\right\}}{\sum_{0 \leq y_1, y_2, y_3 \leq 3} \exp\left\{\sum_{1 \leq j \leq 3} \beta u(y_j) + \theta[I(y_1 = y_2) + 2 \times I(y_2 = y_3)]\right\}}. \text{ The associated}$$

pseudolikelihood entails a multiplication of three terms, each of which corresponds to each individual, is given by

$$pl(y | \theta) = \frac{\exp\{\beta u(y_1 = j) + \theta I(y_1 = y_2)\}}{\sum_{0 \leq y_1 \leq 3} \exp\{\beta u(y_1) + \theta I(y_1 = y_2)\}} \times \frac{\exp\{\beta u(y_2 = j) + \theta I(y_2 = y_3)\}}{\sum_{0 \leq y_2 \leq 3} \exp\{\beta u(y_2) + \theta I(y_2 = y_3)\}}$$

$$\times \frac{\exp\{\beta u(y_3 = j) + \theta I(y_3 = y_2)\}}{\sum_{0 \leq y_3 \leq 3} \exp\{\beta u(y_3) + \theta I(y_3 = y_2)\}}. \text{ The normalizing constant for each component in the}$$

pseudolikelihood is much easier to compute, as it only involves a summation over four terms (i.e., four choice options) as opposed to 64 terms (i.e., all possible values of $y = 4^3$) in the MRF specification.

The pseudolikelihood function is not equivalent to its associated likelihood function and does not sum up to 1 except for the trivial case where all choices are independent of each other. However, the maximum pseudolikelihood estimator is proven to be consistent as n goes to infinity under certain conditions (Geman and Graffigne 1987). Yet, the efficiency of a maximum pseudolikelihood estimator as compared to that of a maximum likelihood estimator is not well established. With a pseudolikelihood¹, traditional Markov chain Monte Carlo (MCMC) procedures can then be implemented on the posterior distribution defined by a pseudolikelihood,

$$p_{PL}(\theta | y) \propto \pi(\theta) pl(y | \theta).$$

¹ Before replacing a likelihood with a pseudolikelihood in the derivation of a posterior distribution, one needs to check whether the pseudolikelihood is proper (i.e., the integral or sum is finite) to ensure that the posterior is also proper.

Approximate Sampling Approach

If we opt to work with the correct likelihood function and carry out a traditional MCMC algorithm, the problem occurs in the Hastings ratio. Note that Gibbs sampling is not an option due to the intractable $Z(\theta)$ in the full conditional distribution. Let $f(y|\theta) = p(y|\theta) \times Z(\theta)$ be the unnormalized likelihood. For the Metropolis-Hastings (M-H) algorithm, the Hastings ratio is

given by $\frac{f(y|\theta')Z(\theta)\pi(\theta')q(\theta|\theta')}{f(y|\theta)Z(\theta')\pi(\theta)q(\theta'|\theta)}$, where q is the proposal density. Normalizing constants

cannot be canceled because they involve unknown parameters. Here we introduce a recently developed MCMC algorithm (Wang and Atchade 2010) for its ease of use and good convergence properties. The key ingredient of the algorithm is to use an auxiliary variable, which is defined in the same space as data, to estimate the ratio of the normalizing constants $Z(\theta)/Z(\theta')$.

The algorithm is as follows:

Step 1: Initialize with some arbitrary θ_0 and auxiliary variable x with arbitrary values in the same space as observed choice data y (y itself is a possible value).

Step 2: Generate θ' from a proposal distribution (e.g., random-walk proposal).

Step 3: Choose κ and compute evenly spaced intermediate points between θ and θ' :

$$\theta^l = \theta + \frac{l}{\kappa}(\theta' - \theta), \quad l = 1, \dots, \kappa.$$

Step 4: Update x^k by a M-H step with invariant distribution $P(y|\theta^l)$ from x^{l-1} (so starting from

$x^0 = x$, update x^1 from x^0 , then x^2 from x^1 , ..., x^κ from $x^{\kappa-1}$). Specifically, we

propose a new value x' for x^l by updating only one value of the choice variable. That is,

we randomly pick an element i in y with equal probability $1/n$ (i.e., an equal probability

for each individual), and generate a value for x_i' with equal probability $1/k$ (i.e., an

equal probability for each choice option). Then with probability $\min\left\{1, \frac{P(x'|\theta^l)}{P(x^{l-1}|\theta^l)}\right\}$, set

$x^l = x'$, and do not change x^l otherwise.

Step 5: With probability:

$$(4) \quad \min\left\{1, \frac{f(D|\theta')\pi(\theta')q(\theta|\theta')}{f(D|\theta)\pi(\theta)q(\theta'|\theta)} \exp\left(\frac{1}{\kappa} \sum_{l=1}^{\kappa} (\theta - \theta')^T \frac{\partial f(y|\theta)}{\partial \theta} \Big|_{\theta=\theta', y=x^l}\right)\right\},$$

we set $\theta = \theta'$ and $x = x^\kappa$; otherwise do not change θ or x .

Step 6: Repeat Steps 2 to 5 until convergence.

If θ is a single parameter, the partial derivative becomes a derivative. Observe that the ratio of $Z(\theta)/Z(\theta')$ is approximated by the quantity inside the exponent, which is also known as path sampling identity (Gelman and Meng 1998), and hence comes the name “approximate sampling.” The κ value is usually chosen to grow with the iteration number, e.g., for the i -th iteration $\kappa = 10 \log(i) + 5$. The larger the κ is, the better estimate of $Z(\theta)/Z(\theta')$ we obtain (i.e., the algorithm generate a sample from the correct posterior density $p(\theta|y)$ as both κ and the number of iteration goes to infinity). If the distance between θ and θ' is large, a bigger κ is desired to produce good estimates.

Discrete-Time Markov Chain for Complete Data

For complete data, we have information on both choices and the choices' sequence. We propose a discrete-time Markov chain to take advantage of the extra information, which can capture ‘who’ makes the choice in addition to ‘which’ choice is made. Following Koskinen and Snijders (2007), we assume that each time at most one member may change the choice (i.e., no

multiple changes are allowed simultaneously). This assumption simplifies the modeling as we do not need to handle the possible dependence of the transitions made at the same time. The state of the Markov chain consists of two parts: who intends to change (individual who makes a choice for the first time changes the choice from 0 to a different value) the choice and everyone's choice. Again, we emphasize that the state of the Markov chain involves all members' choices rather than a single person's due to the interdependence. We denote 'who intends to change the choice' by c^t , which takes value from 1 to n . The state at time point t is $s^t = (c^t, y_1^t, \dots, y_n^t)$. The first-order Markovian property dictates that:

$$(5) \quad p(s^{t+1} | s^1, \dots, s^t, \theta) = p(s^{t+1} | s^t, \theta).$$

Under the assumption that only one transition is allowed, only one person's choice $y_{c^{t+1}}^{t+1}$ can be different from $y_{c^{t+1}}^t$, while all other choices remain the same $y_j^{t+1} = y_j^t$ for $j \neq c^t$. Then we have:

$$(6) \quad \begin{aligned} p(s^{t+1} | s^t, \theta) &= P(c^{t+1} = i, y_i^{t+1} | c^t, y^t, \theta) I(y_{-i}^{t+1} = y_{-i}^t) \\ &= p(c^{t+1} = i | c^t, y^t, \theta) p(y_i^{t+1} | c^{t+1} = i, c^t, y^t, \theta) I(y_{-i}^{t+1} = y_{-i}^t) \end{aligned}$$

The likelihood is $p(s^1, \dots, s^T | s^0, \theta) = \prod_{t=0}^{T-1} p(s^{t+1} | s^t, \theta)$, where y^0 and c^0 are to be specified

later. There is great flexibility in specifying the choice part $P(y_i^{t+1} | c^{t+1} = i, c^t, y^t, \theta)$ of the transition probability. For multinomial choices, a multinomial logit model can be used to fit the choice part. Researchers can postulate any probability form of interest for $P(c^{t+1} = i | c^t, y^t, \theta)$.

We include $I(y_{-i}^{t+1} = y_{-i}^t)$ for completion to assure that only one choice can be made at a time.

We make several remarks about the model. First, a transition/change does not necessarily mean making a different choice. We allow each individual i to keep the same choice (i.e., an individual can log on to the website but then log out keeping the same choice he or she has chosen before in our experiment). Second, modeling everyone's choice together preserves the interdependence within the system. More importantly, we can model who will make the next choice decision. It uncovers the mechanism of the preference dynamics, and thus adds predictive power to the model. Finally, it is easier to identify asymmetric dependence structure in a discrete-time Markov chain model, as in each transition only one person's choice is influenced by others', and thereby statistics corresponding to the parameters for asymmetry always have different values. Despite the differences in the formulation and the type of data they handle, the discrete-time Markov chain and the MRF are closely connected. We provide proof in the Appendix that under certain conditions, the discrete-time Markov chain will converge to the MRF.

EXPERIMENTAL DESIGN AND DATA COLLECTION

We designed a field experiment that facilitates us to collect longitudinal choice data where choices made in a social network are recorded in real time (complete data). As a result, we at the same time obtain cross-sectional (snap-shot) data that are the realization of the complete data. There are several design issues we consider. First, we obtain data from a relatively well-connected social network. Our network involves the undergraduate students who took an introduction to marketing class in the fall of 2009 at a Midwest university. The majority of students knew each other. We expect this network to be denser than that of a randomly selected group of students, and close to a friendship network in reality.

Second, our experimental design circumvents possible confounds from homophily or endogenous group formation (i.e., individuals with similar preferences are more likely to be

friends). In order to tease the influence of others' choices in a network on an individual's choice apart from homophily, we need to control for each individual's intrinsic preference. As such, we measured each participant's preferences for product attributes using a conjoint choice experiment at the same time we collected network information. Without the ability to observe longitudinal choice decisions within-subject and include individual fixed effects (Nair, Manchanda, and Bhatia 2009) or random effects (Hartmann 2009; Yang et al. 2009) in the model, or to measure initial preferences, researchers may include individual covariates as proxies of intrinsic preferences.

Lastly, we choose two products of interest to students, a bundle of university sports paraphernalia and a Bluetooth wireless headset, to motivate them to reveal their true preferences. Our participants also had opportunities to win products of their choices (both the paraphernalia bundle and Bluetooth wireless headset) as an additional incentive to participate in our study. Our paraphernalia bundle contains a jacket, a t-shirt, and a hat with the university logo. We ran the experiment during the football season to further enhance the participants' interests.

Our data collection procedure consisted of three phases. Phase 1 was recruitment. Participants completed two conjoint choice tasks, one for the university paraphernalia and the other for a Bluetooth wireless headset. They also answered a series of social network questions, along with some demographic questions. The purpose was to obtain participants' initial product preferences. Effects of choice interdependence can be identified if participants' preferences for product attributes, as reflected through their choices, change as they observe other participants' choices. Each participant completed twelve conjoint tasks, each involving four options, for each product. We also included the fifth option of "an ear-plugged wired headset at \$29" for the Bluetooth wireless headset conjoint experiment (Figure 2).

We included four attributes for the paraphernalia bundle: jacket style (four levels: style 1-blue, style 1-grey, style 2-blue and style 2-grey); t-shirt color (two levels: blue and white); hat style (two levels: style 1 and 2); and price (four levels: \$39, \$49, \$59 and \$69). We included six attributes for the Bluetooth wireless headset: brand (two levels: Motorola and Plantronics); color

(two levels: black and silver); weight (two levels: 8 and 13 grams); talk time per battery charge (two levels: 5 and 8 hours); noise cancellation (two levels: yes and no); and price (four levels: \$49, \$59, \$69 and \$79). For each product, a blocked design involving ten sets of twelve quadruples were created using SAS OPTEX. Participants were randomly assigned to one of the ten sets.

For network information, we first asked participants to nominate ten students in the class s/he had interacted with most frequently². Next, participants were asked to rate these people with regard to the friendship closeness and the likelihood of sharing purchase opinions. These relationship weights were used to examine which had a stronger effect on final choices.³ To ensure privacy, we provided the subjects with an option to indicate to which nominees they were unwilling to disclose their preferences.

At the end, we asked participants to enter two online raffles. We collected demographic information such as name, gender, most frequently used email, time spent on email every day, confidence in apparel taste, knowledge about Bluetooth headset, and interest in winning the products for students who agreed to participate in the raffles.

In Phase 2, we launched the online raffle for the university paraphernalia bundle (Figure 2). All participants received emails with the instruction to enter the raffle at the same time. A link included in the email directed them to a website, where they were asked to select a paraphernalia bundle they wish to receive should they win the raffle. We provided participants with four paraphernalia options describing by the same attribute levels as shown to them in Phase 1⁴. The four options were chosen according to the results from a pre-test with a similar pool of participants. We expected the four options to be relatively close with regard to their utilities.

Figure 2

Raffle Website for Bundle of Sports Paraphernalia

² To facilitate the nomination process, our JAVA program has a built-in search tool. Participants could easily locate the nominees' names by typing part of their first or last names. The search tool helps us avoid typos and solves the problem that participants may not necessarily remember the full name of all the nominees.

³ We did not find any particular type of network that dominates the rest in the model comparison.

⁴ We omitted price, as including price may have prompted students to choose the most expensive bundle.

Below are four Michigan paraphernalia bundle options available for your raffle prize (in the event that you win). To the right of each bundle, we have indicated the choices of some of your classmates as well as the choice percentages for all students that have made their selections. Please select one of the options (A, B, C, or D) for your raffle prize by clicking on it.

Prize Bundle	SOME of your classmates who chose this bundle	Among ALL students who responded, the percentage that chose this bundle	Among MALE students who responded, the percentage that chose this bundle	Among FEMALE students who responded, the percentage that chose this bundle
A Jacket: Blue Style 1 T-shirt: White Hat: Style 1	Friend 1	31%	38%	20%
B Jacket: Grey Style 1 T-shirt: White Hat: Style 2		39%	28%	59%
C Jacket: Blue Style 2 T-shirt: Blue Hat: Style 2		13%	13%	11%
D Jacket: Grey Style 2 T-shirt: Blue Hat: Style 1	Friend 2	17%	21%	10%
	Among SOME of your classmates	Among ALL students	Among MALE students	Among FEMALE students
Have not yet responded	Friend 3	23%	21%	27%

Note: If you wish to change your selection later, you can come back and submit a new one anytime before Nov 15th, 11:59 PM. Only the last submission will enter the raffle.

Please click this button to confirm your selection.

If you don't want to make your selection at this time, please click this button to exit and come back later.

To facilitate the propagation of social influence, we showed the participants choices made by their “friends” (micro-level relation), as well as the aggregate proportions of participants choosing each option (macro-level relation). The aggregate proportions were computed based on: 1) all participants who had already chosen an option; 2) only female participants (i.e., a reference group) who had already chosen an option; and 3) only male participants who have already chosen an option at a particular time point. The term “friends” here are referred to as other participants who had agreed to share preferences with each participant in Phase 1. Participants could change their selections as often as they wished before the end of the raffle period.

The novelty of the experiment stems from the real-time delivery of friends’ choices

during the raffle period. The procedure works as follows. Every time a participant made or changed a choice, emails were sent out to his or her 'friends.' We recorded everyone's actions. Our database was updated real-time to assure that the website offered up-to-date information. To increase the response rate, we sent a reminder to enter the raffle in the middle of the raffle period.

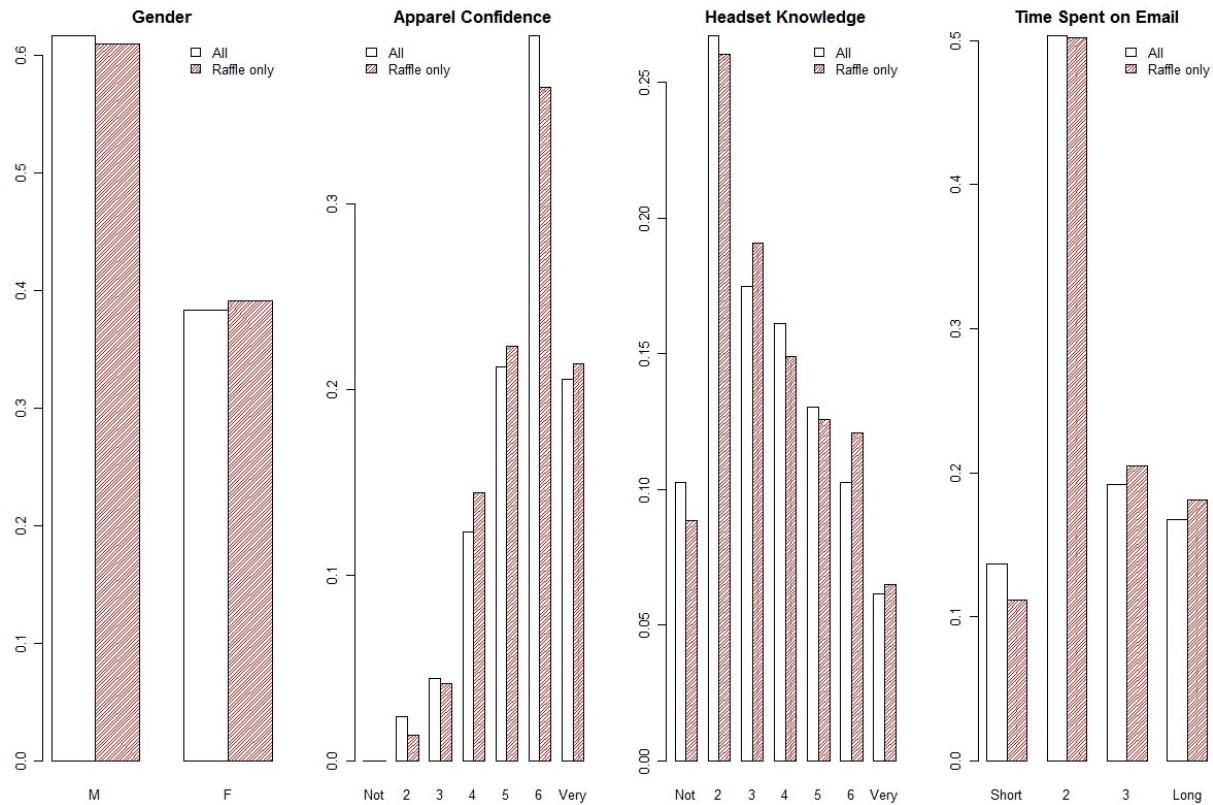
Phase 2 lasted for a week, and then we launched the second raffle for the Bluetooth wireless headset in Phase 3. The set-up of the second raffle is very similar to the first one, except that if participants did not make a selection, the best product they could receive was a wired headset at a much lower price than any of the Bluetooth options. The raffle was closed after a week, and two reminders were sent during this period. Our experimental design makes it possible for us to obtain both snap-shot and complete data. Complete data come from our track of the logon activities and confirmed choice selections. Snap-shot data are all the selections we observed by the time raffles closed.

EMPIRICAL ANALYSIS

Sample Profile

In Phase 1, we have 292 participants in total, among which 215 agree to participate in the raffles. All the analysis hereafter is based on these 215 participants. The demographic profiles of all the participants (292) is very similar to those in the raffle (215) (Figure 4). Thus, the raffle participants are considered a random sample of all the participants from Phase 1.

Figure 3 Demographic Information



Among the 215 raffle participants, 39.1% are female and 18.6% are categorized as ‘experts’ on the Bluetooth headset (we asked participants to rate themselves on a 1-7 scale in Phase 1 how knowledgeable they are about Bluetooth headset and experts are those with ratings higher than 5). More than half of the participants are ‘very interested in winning a paraphernalia bundle’ with a rating of 7 on a 1-7 scale (1 is ‘not interested all at’), and the mean rating is 6.0. For the Bluetooth headset, students’ interests are more evenly distributed, although one-third of the ratings are 7, and the mean rating is 4.5 (on a 1-7 scale).

Social Network

On average, every participant had 3.7 friends. The most popular participants have 12 friends, while 14 participants have no friends. Table 1 provides some network structure statistics. As expected, the density, reciprocity and transitivity are low as we work with a relatively large network.

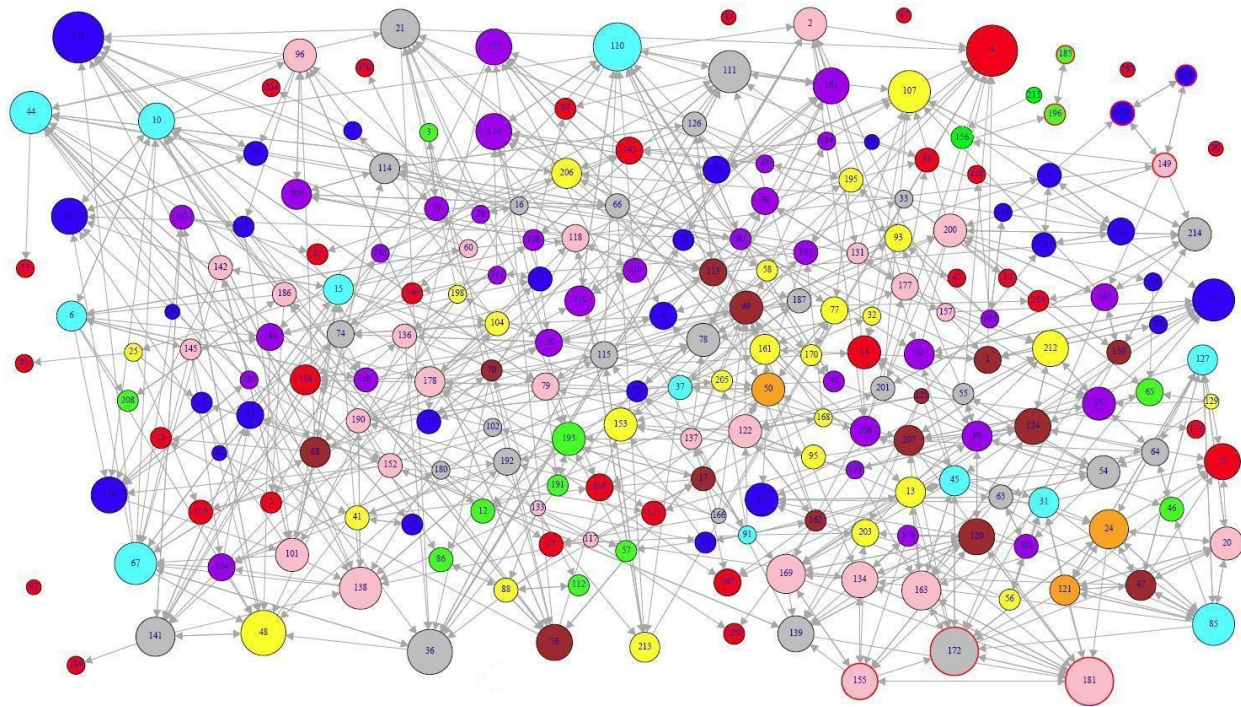
Table 1
Network Summary Statistics

Effects	Network Statistics and Graph Illustration			Values
Mean in/out degree	$\sum_{1 \leq i, j \leq n} e_{ij} / n$			3.7
Variance of in degree (prestige)	$\text{var}(\sum_{1 \leq j \leq n} e_{ji})$	i	$\longrightarrow$	7.0
Variance of out degree	$\text{var}(\sum_{1 \leq j \leq n} e_{ij})$	i	$\longrightarrow$	6.0
Percentage of isolates	$\#(\sum_{1 \leq j \leq n} e_{ji} = \sum_{1 \leq j \leq n} e_{ij} = 0) / n$	i		2.3%
Percentage of senders	$\#(\sum_{1 \leq j \leq n} e_{ji} = 0, \sum_{1 \leq j \leq n} e_{ij} > 0) / n$	i	$\begin{array}{c} \longleftarrow \\ \longrightarrow \end{array}$	4.2%
Percentage of receivers	$\#(\sum_{1 \leq j \leq n} e_{ji} > 0, \sum_{1 \leq j \leq n} e_{ij} = 0) / n$		$\begin{array}{c} \longrightarrow \\ \longleftarrow \end{array}$	12.6%
Density	$\sum_{1 \leq i, j \leq n} e_{ij} / n(n-1)$			1.7%
Percentage of reciprocal connections	$\sum_{1 \leq i < j \leq n} 2e_{ij}e_{ji} / n(n-1)$	i j	$\begin{array}{c} \longrightarrow \\ \longleftarrow \end{array}$	.73%
Percentage of transitive triads (ordered)	$\sum_{1 \leq i, j, i' \leq n, i \neq j \neq i'} e_{ij}e_{ji'}e_{ii'} / n(n-1)(n-2)$	i i'	$\begin{array}{c} j \\ \nearrow \searrow \\ \longrightarrow \end{array}$	.007%

* $e_{ij} = 1$ if i sends information to j , 0 otherwise.

Figure 5 depicts the directed social network among the raffle participants. Each node represents a participant. An arrow represents a connection. A node with larger size indicates a more popular participant with more friends (i.e., receivers with arrows in), and nodes in the same color have the same number of arrows out (i.e., senders). Notice that there are five socially peripheral participants with no arrow in or out. Among the 628 pairs of connected participants (the same two individuals are only counted as one pair), 26.6% of them have arrows both in and out present (e.g., they have reciprocal connections).

Figure 5
Network Plot



Raffle Choice Summary

For the paraphernalia bundle, 167 participants visit the website, and 62.3% of them are male, which is similar to the proportion of males among the 215 raffle participants. Among the 25 participants who visit the site multiple times, 17 did not change their choices in all visits, whereas the remaining eight first did not choose any option but then pick an option later. The percentages of people choosing options 1 to 4 and not choosing any of the options are 30.5%, 38.9%, 9.0%, 16.8%, and 4.8%, respectively.

For the Bluetooth headset, 97 participants visit the website, and 25.8% are 'experts,' which is higher than the percentage of experts among the 215 enrolled (18.6%). The main reason could be that experts are more interested in winning a Bluetooth headset than novices, as the mean ratings of interest are 5.9 and 4.2 for experts and novices, respectively. Among the 97 participants, 92 also logged on to the bundle raffle website. Thirty-four participants visit the site multiple times, and 28 of them maintain the same choices in all the visits. The remaining 6 first did not choose an option but later choose one. The percentages of people choosing options 1 to 4 and not choosing any of the options are 14.4%, 3.1%, 74.2%, 2.1%, and 6.2%, respectively.

Model Estimation

We fit three models to the experiment data: (1) a baseline hierarchical Bayes logit model; (2) the same baseline model for Phase 1 data and also fit the MRF model with multiple relations to the raffle data (from Phases 2 and 3); and (3) model 2 except that we fit the discrete-time Markov chain model to the data instead of the MRF model.

Hierarchical Bayes Model

We use the hierarchical Bayes logit model (Allenby and Rossi 1999) to fit the conjoint data from Phase 1. In the absence social network information, the best guess of a participant's choice in the raffle is based on his or her estimates of initial preferences for different product attributes. The model specification is as follows.

Participant i 's utility of option j is $u_{ij}^I = \sum_{p=1}^P \beta_{ip} x_{pj}^I + \varepsilon_{ij}^I$, where β_{ip} captures individual i 's

preference for the p -th product attribute, x_{pj}^I indicates the p -th attribute's value for product j , ε_{ij}^I is an error term following type I extreme distribution, and the superscript I means it is related to initial preferences. All the attributes are dummy-coded including the price. In the Bluetooth

headset case, the utility for the wired headset is $u_{i0}^I = 0$, consequently those for Bluetooth wireless

headsets have an intercept term β_0 . Hereafter, we mainly use the model for the paraphernalia

bundle as an illustration. Let $n = 215$, $\beta_i = (\beta_{i1}, \dots, \beta_{iP})$, $\beta = (\beta_1, \dots, \beta_n)$, i 's choices in 12

conjoint tasks $y_i^I = (y_{i1}^I, \dots, y_{i12}^I)$, the corresponding attributes for option j in q -th task

$x_{iq,j}^I = (x_{iq,1j}^I, \dots, x_{iq,Pj}^I)$, $y^I = (y_1^I, \dots, y_n^I)$, $x^I = (x_{11,1}^I, \dots, x_{n12,4}^I)$, and P be the number of attributes,

then the likelihood is given by:

$$(7) \quad p(y^I | x^I, \beta) = \prod_{i=1}^n \prod_{q=1}^{12} \frac{\exp\{\beta_i^T x_{iq,y_{iq}^I}^I\}}{\sum_{j=1}^4 \exp\{\beta_i^T x_{iq,j}^I\}}.$$

We follow the conventional prior setting of a hierarchical Bayes model:

$b \sim N(0, 100I_{P \times P})$, $V_\beta \sim IW(P+2, I_{P \times P})$, and $\beta_i \sim N(b, V_\beta)$, where $I_{P \times P}$ is a $P \times P$ identity matrix. We carry out the Metropolis-within-Gibbs algorithm. According to the trace plots and auto-correlation plots with 10,000 iterations, the Markov chain has converged and has good mixing. The estimation and inference are based on the every tenth posterior draws from the last 8,000.

Markov Random Field with Multiple Relations

We fit the MRF with multiple relations to snap-shot data, which correspond to participants' final choices when the raffle is closed. We only model the choices of participants who at least log on to the raffle website, and treat the rest as if they decline the raffle in Phase 1. The fact that they did not choose any option will enter the model as a covariate.

Apparently, we have a directed network of sharing choice decisions. We believe the conditional probability of one's choice depends on both senders and receivers (i.e., two consumers are neighbors as long as at least one of them shares choice decisions with the other). When specifying the potentials (U_c), we focus on cliques of size one and two. Potentials related to cliques of size one capture the effects of each individual's initial product selection. In case of not choosing any option, the attributes of this option are defined as the mean values of four options because we inform participants that if they do not make a raffle choice, they will receive a randomly chosen option as their prize. In addition, we include an intercept accounting for other rationale behind this "no-choice" option and treat it as an 'outside good' utility.

For cliques of size two, i.e., any sender and receiver, we expect the sender to influence the receiver. This is a micro-level relationship. To account for the influence of this relationship, we sum the number of an individual's senders who make the same choice. To control for a

different number of friends, we normalize the summation by the number of friends (i.e., senders) each receiver has. Thus, this binary network assigns a weight of 1 to every connection.

On the raffle website, we also make available the summary information on the choices made among all participants. We use the information about choices of the reference group to capture the influence from the macro-level relationship. The overall summary is not used due to its high correlation with the reference group summary. To compute potential of cliques related to this macro-level of relationship, we make participants in the same reference group neighbors. Within each clique (e.g., male vs. female), we expect that participants tend to choose the option that more members in their clique have chosen.

We further assume that not choosing any option is not affected by any network effect. Let $m = 167$, $y = (y_1, \dots, y_m)$ representing 167 participants' final choice decisions, 0 be “no-choice” option, $x = (x_1, \dots, x_4)$ with x_j the vector of attributes of option j in the raffle, $\bar{x} = \sum_{j=1}^4 x_j / 4$ be the mean values for all the attributes, γ be a vector of parameters associated with the network effect, and $W = (w_{ij})$ be the matrix of binary directed network with $w_{ij} = 1$ if person i sends information to j and 0 otherwise. g_i is i 's gender in bundle raffle and if i is an expert in the headset raffle.

The specification of the MRF with multiple relationships described above is:

$$(8.a) \quad p(y | x, \beta, \gamma, g, W) = \exp \left\{ \sum_{i=1}^m [I(y_i = 0)(\gamma_0 + \gamma_1 \beta_i^T \bar{x}) + I(y_i > 0) \gamma_1 \beta_i^T x_{y_i}] + \right. \\ \left. \sum_{1 \leq i, j \leq m} \gamma_2 w_{ij} I(y_j > 0) \frac{I(y_j = y_i)}{\sum_{i'=1}^n w_{i'j}} + \sum_{h=1}^2 \gamma_3 \sum_{i=1}^m I(y_i > 0) \frac{\sum_{j=1, \neq i}^m I(y_i = y_j, g_i = g_j = h)}{\sum_{j=1, \neq i}^m I(y_j > 0, g_i = g_j = h)} \right\} / Z(\beta, \gamma)$$

where γ_0 is the intercept for the “no-choice” option, γ_1 adjusts the magnitude of the initial

preferences (i.e., a scale parameter), γ_2 is the local friends' influence (i.e., micro-level relationship), and γ_3 is the reference group influence (i.e., macro-level relationship). We also

define $\frac{0}{0} = 0$ for all the ratios. The first summation involves cliques of size one, the second summation involves cliques of size two that capture the micro-level relationship, and the last summation involves cliques of size two that capture the macro-level relationship. Note that the denominator of the last summation is the number of participants who chose a particular option. For the Bluetooth headset, the likelihood is different, as our main focus is on the asymmetric dependence between students who are knowledgeable (i.e., experts) and not knowledgeable (i.e., novices) about Bluetooth. Also, selection of 0 is modeled differently as $-\gamma_0\beta_{i0}$, since it represents a wired headset at \$19.99. The specification is:

$$(8.b) \quad p(y|x, \beta, \gamma, g, W) = \exp\left\{\sum_{i=1}^m [-I(y_i = 0)\gamma_0\beta_{i0} + I(y_i > 0)\gamma_1\beta_i^T x_{y_i}]\right\} +$$

$$\sum_{\substack{i \text{ is expert} \\ j \text{ is novice}}} \gamma_2 w_{ij} I(y_j > 0) \frac{I(y_j = y_i)}{\sum_{i'=1}^n w_{i'j}} + \sum_{\substack{i \text{ is novice} \\ j \text{ is expert}}} \gamma_3 w_{ij} I(y_j > 0) \frac{I(y_j = y_i)}{\sum_{i'=1}^n w_{i'j}} \} / Z(\beta, \gamma).$$

Thus γ_2 reflects novice's choice dependence on expert's, and γ_3 reflects expert's on novice's.

We include all 215 people's data in Phase 1 to get better estimates of initial preferences. The full likelihood contains both conjoint and raffle final choices,

$$p(y, y^I | x, x^I, \beta, \gamma, g, W) = p(y | x, \beta, \gamma, g, W) p(y^I | x^I, \beta), \text{ where } p(y^I | x^I, \beta) \text{ is defined in (8).}$$

The prior for the new parameters is $\gamma \sim N(0, 100I_{4 \times 4})$, the rest is the same as (9).

We have two approaches to fit the MRF with multiple relations. For the approximate

sampling approach, we sample from the posterior distribution derived from the full likelihood using the algorithm described above (see the Appendix for details). The pseudolikelihood approach involves replacing the complicated likelihood of MRF with multiple relations with the corresponding pseudolikelihood. With the same prior as above, we can obtain posterior draws from $p_{PL}(\theta | y)$ by running the Metropolis-within-Gibbs algorithm.

Table 2 gives the posterior mean for the parameter estimates. Ten thousands iterations are carried out in both the algorithms, with $\kappa = 10 \log(i+1) + 100$ in the approximate sampling. Estimation and inference are based on the every tenth posterior draws from the last 8,000. To perform a hypothesis test, we calculate the probability of a parameter ≤ 0 by counting the number of draws above 0.

We see in Table 2 that both the pseudolikelihood and approximate sampling methods give very close results. In the bundle raffle, both local friends' influence and the social norm manifest strong effects on the final. In the Bluetooth headset raffle, we see effect from both experts and novices, with a large magnitude for experts' influence.

Table 2
Parameter Estimates

Parameter		Markov Random Graph		Discrete-time Markov chain
		Pseudolikelihood	Approximate Sampling	
Bundle	Initial Preference (β):			
	Style 1 blue jacket ¹	1.02**	.970**	1.02**
	Style 1 grey jacket	1.42**	1.41**	1.47**
	Style 2 blue jacket	-.261	-.285* [†]	-.274* [†]
	Blue t-shirt ²	.305**	.307**	.294**
	Style 1 hat ³	.255**	.287**	.232*
	\$39 ⁴	3.16**	3.11**	3.14**
	\$49	2.45**	2.42**	2.45**
	\$59	1.14**	.993**	1.11**
	Social Influence:			
	Outside good (α_0)	-.693** [†]	-.755** [†]	.200
	Scale for initial preference (α_1)	.465**	.480**	.469**
	Local friends' influence (α_2)	.623**	.526**	.945**

	Social norm effect (β_3)	1.23**	1.24**	2.09**
	Weighted friends' effect on who will change	n/a	n/a	.093*
Bluetooth	Initial Preference (β):			
	Ear-plug headset at \$29 ($-\beta_0$)	-1.36	-1.22** [†]	-2.53** [†]
	Motorola ⁵	.802**	.777**	.726**
	Black ⁶	-.105	-.127	-.136
	8 grams ⁷	1.54**	1.36**	1.63**
	8 hours ⁸	1.64**	1.47**	1.62**
	Noise cancellation	1.80**	1.82**	1.93**
	\$49 ⁹	4.29**	3.86**	4.20**
	\$59	2.40**	2.16**	2.28**
	\$69	.916**	.629**	.410**
	Social Influence:			
	Outside good (β_0)	-.020	-.053	-.045
	Scale for initial preference (β_1)	.353**	.371**	.357**
	Local friends' influence (experts to novices; β_2)	4.31**	4.68**	7.10**
	Local friends' influence (novices to experts; β_3)	2.04*	3.66**	5.91**
	Weighted friends' effect on who will change	n/a	n/a	-.041

* significant at .99 level, ** significant at .95 level, [†] test parameter greater than or equal to 0 reference level: ¹ style 2 grey jacket, ² white t-shirt, ³ style 2 hat, ⁴ \$69, ⁵ Plantronics, ⁶ silver, ⁷ 13 grams, ⁸ 5 hours, ⁹ \$79.

Discrete-time Markov Chain

We model the choices made by the participants using a similar specification to the MRF model (equations 8a and 8b). To model who intends to change a choice at time point $t+1$, c^{t+1} ,

we use a logit model with utility $u_i^{t+1} = \alpha e_i \frac{\sum_{j=1}^m w_{ji} I(y_j^t > 0)}{\sum_{k=1}^n w_{ji}} + \varepsilon_i^{t+1}$, where α is the parameter, e_i

is a weight taking values from 1 to 4 based on each individual's i rating of his or her frequency of email usage, the error term ε_i^{t+1} follows a type I extreme value distribution. w_{ij} and g_i are defined as above. The intuition is that participants likely make their choice if more of their friends have made a selection, and if they check emails more frequently. Its likelihood is

$$p(c^{t+1} = i | c^t, y^t, x, \beta, \gamma, g, W) = \frac{\exp\{u_{c^{t+1}}^{t+1}\}}{\sum_{i=1}^n \exp\{u_i^{t+1}\}}.$$

Let $c^{t+1} = i$, in the bundle raffle each $p(y_i^{t+1} | c^{t+1} = i, c^t, y^t, x, \beta, \gamma, g, W)$ has a logit form with:

$$(9.a) \quad u_{iy_i^{t+1}}^{t+1} = I(y_i^{t+1} = 0)(\gamma_0 + \gamma_1 \beta_i^T \bar{x}) + I(y_i^{t+1} > 0) \gamma_1 \beta_i^T x_{y_i} + \gamma_2 I(y_i^{t+1} > 0) \frac{\sum_{j=1}^m w_{ji} I(y_i^{t+1} = y_j^t)}{\sum_{j'=1}^n w_{j'i}} +$$

$$\gamma_3 I(y_i^{t+1} > 0) \frac{\sum_{j=1, \neq i}^m I(y_i^{t+1} = y_j^t, g_i = g_j)}{\sum_{j=1, \neq i}^m I(y_j^t > 0, g_i = g_j)} + \varepsilon_{iy_i^{t+1}}^{t+1}.$$

In the headset raffle, $p(y_i^{t+1} | c^{t+1} = i, c^t, y^t, x, \beta, \gamma, g, W)$ is:

$$(9.b) \quad u_{iy_i^{t+1}}^{t+1} = -I(y_i^{t+1} = 0) \gamma_0 \beta_{i0} + I(y_i^{t+1} > 0) \gamma_1 \beta_i^T x_{y_i} +$$

$$\gamma_2 I(y_i^{t+1} > 0, i \text{ is novice}) \frac{\sum_{j=1}^m w_{ji} I(y_i^{t+1} = y_j^t, j \text{ is expert})}{\sum_{j'=1}^n w_{j'i}} +$$

$$\gamma_3 I(y_i^{t+1} > 0, i \text{ is expert}) \frac{\sum_{j=1}^m w_{ji} I(y_i^{t+1} = y_j^t, j \text{ is novice})}{\sum_{j'=1}^n w_{j'i}} + \varepsilon_{iy_i^{t+1}}^{t+1}.$$

The full likelihood is: (10)

$$p(s^1, \dots, s^T | s^0, x, \beta, \gamma, g, W) = \prod_{t=0}^{T-1} \frac{\exp\{u_{c^{t+1}}^{t+1}\}}{\sum_{i=1}^n \exp\{u_i^{t+1}\}} \frac{\exp\{u_{c^{t+1} y_{c^{t+1}}}^{t+1}\}}{\sum_{k=I(y_{c^{t+1}}^t=0)}^4 \exp\{u_{c^{t+1} k}^{t+1}\}} I(y_{-c^{t+1}}^{t+1} = y_{-c^{t+1}}^t).$$

Unlike in MRF(MN), people who selected an option last time can no longer change back to ‘not

select an option,' hence $k = I(y_{c^{t+1}}^t = 0)$. Let $c^0 = 0$, meaning nobody intends to change before the raffle starts, and y^0 is a vector of 0, indicating that everyone starts without any selection. Note that the likelihood of y^{t+1} is very similar to the pseudolikelihood, but the statistic values are different. That is, statistics in the discrete-time Markov chain are computed based on the previous time point, whereas those in pseudolikelihood are computed based on choices observed at the last time point.

We ran the Metropolis-within-Gibbs algorithm for 10,000 iterations (see the Appendix), and use every tenth draw from the last 8,000 to do estimation and inference (Table 2). Since complete data contain more information than the snap-shot data, the magnitude of the parameters are not really comparable, although there are commonalities. All the parameters related to individual intrinsic preferences (i.e., initial preferences) for different product attributes appear reasonable. The scale parameter for initial preference is significant for both categories suggesting that initial preferences still play a significant role in determine participants' choice for products in the raffles. Using data from the paraphernalia bundle category, we find participants rely on the choices of others both at the micro and macro levels in making their decisions. Using data from the Bluetooth category, we account for potential asymmetry in choice interdependence between experts and novices. As expected, experts have stronger influence on novices' choices than vice versa. In the discrete-time Markov choice model, we also model "who intends to change a choice" and find that for the paraphernalia bundle category, participants are more likely to make choices if more of their friends have made choices.

Model Comparison

In this section, we compare the three models described above using in-sample and out-of-sample hit rates. Hit rate is computed as the mean percentage of correctly predicted selections among all respondents. We opt to use hit rates instead of traditional fit measures such as log marginal density and DIC because these traditional measures require the calculation of the normalizing

constant of the Markov random field, which is computationally expensive. To make the results more comparable across models, in the discrete-time Markov chain model the two hit rate measures are computed only for final choices, and in prediction, we condition on who intends to change the choice. We randomly pick 10% of the respondents and exclude their final choices for out-of-sample fit.

Making predictions with the baseline model is straightforward. Since people's selections are treated as independent, we can draw each prediction from its own posterior predictive

probability $P(y_i^* | D) = \int P(y_i^* | \theta) p(\theta | D) d\theta \approx \sum_l P(y_i^* | \theta_l)$, where θ_l is the posterior draw. The

utility for the choice of 0 is defined as $x\bar{\beta}_i$ in bundle raffle, and $-\beta_{i0}$ in headset raffle.

Prediction in the remaining models is quite different from the traditional in-sample or out-of-sample prediction for independent data. Interdependent choices should to be predicted jointly rather than separately. When data are left out, we cannot fit the model only on the remaining data as it will destroy the dependence structure in the network. Nor can we just assign a selection of 0 to out-of-sample people. Prediction strategies for the dependent data are below.

For in-sample prediction, we need to draw from $P(y^* | D) = \int P(y^* | \theta) p(\theta | D) d\theta$. It usually takes less computing time to generate y^* within the model fitting process than from each posterior $P(y^* | \theta_l)$, although for a Markov chain model a simpler procedure exists. More specifically, we draw parameters along with the predictions from

$P(\theta, y^* | D) \propto P(D | \theta) P(y^* | \theta) \pi(\theta)$. More details can be found in the Appendix. To make the comparison more comparable across models, we fixed the order in the Markov chain for both in-sample and out-of-sample predictions.

For out-of-sample fitting and prediction, we fit the model with all the data and treat the out-of-sample data as missing values. In other words, we draw parameters and predictions

together from $P(\theta, y_{out}^* | D_{in}) \propto P(D_{in}, y_{out}^* | \theta)\pi(\theta)$, where y_{out}^* is the out-of-sample prediction and D_{in} is the in-sample data. It is not hard to implement this in a Bayesian framework (see the Appendix).

Results are shown in Table 3. In all the cases, the MRF and discrete-time Markov chain models perform better than the baseline model, especially for the Bluetooth category. These results suggest the importance of understanding choice interdependence in a social network. We expect choice interdependence to play a more important role for the Bluetooth category because students are less likely to have strong preference for such a high-tech product, as compared to a paraphernalia bundle, and hence rely on their peers' choices to make their own decisions. Both pseudolikelihood and approximate sampling approaches appear to perform equally well. We initially expected the MRF model estimated by the approximating approach to perform better than that by the pseudolikelihood. However, recall that prior research has proven the consistency of pseudolikelihood estimates as the sample size goes to infinity. Working with a relative large network in our field experiment may not help showcase the benefit of using the approximating sampling method.

Table 3
Model Comparison

Measure		Markov Random Field (Model 2)		Discrete-time Markov Chain (Model 3)	Baseline (Model 1)
		Pseudolikelihood	Approximate Sampling		
Bundle	In-sample hit rate	41.0%	42.4%	40.0%	39.7%
	Out-of-sample hit rate	39.8%	40.0%	38.9%	38.2%
Bluetooth	In-sample hit rate	39.7%	36.9%	37.3%	28.4%
	Out-of-sample hit rate	37.6%	38.3%	40.8%	26.1%

To further compare the performance of pseudolikelihood and approximate sampling approaches, we carry out the following simulation example. Suppose students in a classroom are

given two different coupons to choose. We define a student's friends as the people sitting in front of and behind him/her, and sitting to his/her left and right. We want to see if students tend to choose the same or different coupons as their friends. The model formulation (known as the Ising model) is as follows. Suppose there are N rows with N students in each row. The j^{th} student in i^{th} row makes a choice of either coupon 1 or 2, i.e., $x_{ij} = \{-1, 1\}$. Let $X = (x_{ij})_{1 \leq i, j \leq N}$, the likelihood is

$$(10) \quad P(X | \theta) = \exp \left\{ \theta \left(\sum_{i=1}^N \sum_{j=1}^{N-1} x_{ij} x_{i,j+1} + \sum_{i=1}^{N-1} \sum_{j=1}^N x_{ij} x_{i+1,j} \right) \right\} / Z(\theta).$$

We simulate the data for $N=5, 10, 20$ corresponding to 25, 100 and 400 students, and $\theta = .25$. Twenty percent of the data are randomly taken out of the sample, and we compare the out-of-sample hit rates of the two methods. The results are presented in Table 4. In all cases, approximate sampling performs better, but as network size increases, the difference between the hit rates of two methods decreases. Hence, when dealing with a large network, due to computational consideration, we would recommend using the pseudolikelihood approach. But for a small and moderate-sized network, the approximate sampling approach provides more accurate predictions.

Table 4
Simulation Results Out-of-sample Hit Rate

N^2	Pseudolikelihood	Approximate Sampling
25	56.1%	62.3%
100	59.9%	62.5%
400	57.3%	58.6%

DISCUSSION

Our paper makes theoretical and methodological contributions. Theoretically, we derive sufficient conditions for the discrete-time Markov chain to converge to the Markov random field, and hence show that Markov random field is a natural model that emerges from the Markov

chain dynamic choice process. Methodologically, we propose a modeling framework for examining choice interdependence in a social network. Specifically, we propose a Markov random field which can accommodate multiple relations to model snap-shot data and a discrete-time Markov chain to model complete data. We also demonstrate how one can use either a pseudolikelihood (Besag 1974, 1975) or an approximate sampling (Wang and Atchade (2010)) approach to circumvent the intractable normalizing constant problem in estimating the MRF.

Our results show that controlling for individuals' heterogeneous preferences, our results individuals rely on choices observed both at the micro (i.e., closely-connected friends) and macro (i.e., a reference group) levels in making their decisions. Choice interdependence is shown to play a more important role for products that are not familiar to consumers. We also find asymmetry in choice interdependence across different types of consumers (i.e., expert vs. novice). With regard to model comparison, discrete-time Markov chain provides more information but does not necessarily predict choices better. The approximate sampling approach performs better than the pseudolikelihood approach for small and medium-sized networks. The difference in their performance dissipates as size of the network increases.

Although the goal of our research is not to model coordinated choices, the MRF model can be used to model coordinated choices. With an exception of Yang et al. (2009), extending existing coordinated choice models to accommodate a larger group is practically implausible. For example, Hartmann (2009) relies on the theory of simultaneous-move games (Bresnahan and Reiss 1991) to model simultaneous choice decisions (i.e., coordinated choices) made by multiple players. Yang et al. (2009) discuss the difficulty in applying such an “unconditional approach” in Hartmann et al. (2009) to situations with more than two choice alternatives or two game players. Yang et al. (2009) opt for a “conditional approach” and introduce an auto-model for multinomial choices, which extends an autologistic model (Besag 1974; Moon and Russell 2007; Russell and Peterson 2000) from binary choices to multinomial choices. However, the auto-model proposed in Yang et al. (2009) also encounters the intractable normalizing constant problem if it were to be

applied to a larger-sized network, as opposed to a network of size three used in their paper.

In fact, an auto-model (Besag 1974) is a special case of the Markov random graph (MRF), where the joint distribution of individual i 's decisions $y_i; i=1, \dots, n$, takes the form of

$$p(y_1, \dots, y_n) = \exp\left(\sum_{1 \leq i \leq n} u(y_i) + \sum_{1 \leq i < j \leq n} \beta_{ij} I(y_i = y_j)\right) / z(\beta), \text{ where } y_i \text{ takes nominal values from } 0, \dots,$$

k (to denote different choice options), $u(y_i)$ is the utility function associated with each nominal value, and $z(\cdot)$ is the normalizing constant, which sums over all possible values of

$y = (y_1, \dots, y_n)$. In Yang et al. (2009), under a constraint that if an individual watches TV, he or she watches the same program with others watching TV at the same time (i.e., assume one TV per household), for n people facing k TV programs, there are $(2^n - 1)k + 1$ possible outcomes. For only fifteen people with four TV programs, the normalizing constant already involves the summation of 131,069 terms, and thus is computationally intractable.

The MRF model is also related to autoregressive models proposed to model preference interdependence (Yang and Allenby 2003). Although the predictors in the MRF model we propose are choices of an individual's friends, with little modification the model can also accommodate preferences (together with choice decisions) as the model predictors. Besag (1974) discusses the connection between auto-models (a special case of the MRF) and autoregressive models. Specifically, auto-models have the Markovian property, that is, one's preference only depends on its neighbors' preferences (e.g., closely connected friends' preferences). On the other hand, in autoregressive models one's preference can depend on preferences of even neighbors of an individual's neighbors and neighbors of the neighbors, and so on and so forth. However, if we expand the scope of neighbors in the MRF specification by including everyone who can affect one's choices, autoregressive models can become a special case of MRF models.

APPENDIX

Connection between MRF(MN) and Markov Chain

In spite of the differences on the formulation and the type of data they handle, Markov chain and MRF can be closely connected. The Markov chain in this subsection we refer to here is the marginalized Markov chain made up of choices only, the likelihood of which is

$$P(y^1, \dots, y^T | y^0, \theta) = \sum_{c^0, \dots, c^T} P(s^1, \dots, s^T | s^0, \theta) P(c^0 | y^0, \theta).$$

The intuition is that as the complete data give rise to the snap-shot data, we speculate that under certain conditions, the Markov chain may converge to the MRF as the dynamic process continues. In this section, we give a sufficient condition for this speculation.

Theorem 1: In the Markov chain model, if (i) the choice part of the transition probability (6) has the same form as the conditional probability of MRF(MN) calculated based on (1) (or (2)), i.e., $P(y_i^{t+1} | c^{t+1} = i, c^t, y^t, \theta) = P(y_i | y_{-i} = y_{-i}^t, \theta)$; (ii) c^{t+1} is independent of y^t , i.e.,

$P(c^{t+1} | c^t, y^t, \theta) = P(c^{t+1} | c^t, \theta)$, then the stationary distribution of the Markov chain is given by the MRF(MN) (1) (or (2)). If further (iii) the transition probability of the Markov chain produces an irreducible and aperiodic Markov chain, then its limiting distribution is (1) (or (2)).

A Markov chain is irreducible if any two states can reach each other within finite number of transitions, i.e., for any two states z, z' (can be the same), there exists $t \geq 0$ such that

$P(y^t = z' | y^0 = z) > 0$. A state z is aperiodic if there is no integer $d > 1$ such that

$P(y^t = z | y^0 = z) = 0$ whenever t is not divisible by d . If all states are aperiodic, the Markov chain is aperiodic. For an irreducible Markov chain, we only need to show that one state is aperiodic to obtain this property for the whole chain.

Proof: The key thing is that our Markov chain can be considered as generated from a Gibbs sampler with stationary distribution of (1) (or (2)). We define the sampler as follows. In iteration $t+1$, we randomly pick an i according to $P(c^{t+1} = i | c^t, y^t, \theta)$, and then generate y_i^{t+1} from $P(y_i^{t+1} | y^t, c^{t+1} = i, \theta)$. Our Markov chain can be obtained by this sampler. Condition (i) and (ii) ensures that it is a Gibbs sampler with invariant distribution (1) (or (2)). Condition (iii) guarantees that the limiting distribution of this sampler is the same as its stationary distribution (Ross 1996). Therefore, our Markov chain will converge to (1) (or (2)) as the number of transitions goes to infinity.

Now we will prove the invariant distribution of the Gibbs sampler is $P(y | \theta)$ defined in

(1) (or (2)). We eliminate θ and denote the choices in consecutive two steps by x and y for notation lucidity. All we need to prove is $\sum_x P(x)Q(y|x) = P(y)$, where $Q(y|x)$ is the transition kernel of the sampler. Then we have

$$\begin{aligned}
& \sum_x P(x)Q(y|x) \\
&= \sum_x P(x) \sum_{i=1}^n P(c^{t+1} = i | c^t, x) P(y_i | c^{t+1} = i, c^t, x) I(y_{-i} = x_{-i}) \\
&= \sum_x P(x) \sum_{i=1}^n P(c^{t+1} = i | c^t) P(y_i | x_{-i}) I(y_{-i} = x_{-i}) \\
&= \sum_{i=1}^n P(c^{t+1} = i | c^t) \sum_x P(x_i | x_{-i}) P(x_{-i}) P(y_i | x_{-i}) I(y_{-i} = x_{-i}) \\
&= \sum_{i=1}^n P(c^{t+1} = i | c^t) \sum_x P(x_i | x_{-i}) P(y_i, x_{-i}) I(y_{-i} = x_{-i}) \\
&= \sum_{i=1}^n P(c^{t+1} = i | c^t) \sum_{x_i} P(x_i | y_{-i}) P(y) \\
&= P(y) \sum_{i=1}^n P(c^{t+1} = i, | c^t) \\
&= P(y)
\end{aligned}$$

Remark: Note condition (ii) in the theorem is a sufficient rather than necessary condition. The

necessary condition takes a less intuitive form, which is $\sum_{i=1}^n \sum_{x_i} P(c^{t+1} = i, | c^t, x_i, y_{-i}) P(x_i | y_{-i}) = 1$

. It is not difficult to prove the above conclusion after replacing condition (ii) with this necessary condition.

Estimation Procedures:

Here we use the specification of bundle raffle as an illustration. We assume that the students are sorted so that the first m students logged on to the website.

Hierarchical Bayes Model

The Metropolis-Hasting algorithm (Chib and Greenberg 1995) was used to generate the empirical draws for estimation. Draws for each parameter were obtained as follows:

1) Generate b

$$f(b | \{\beta_i\}, V_\beta) = N \left(\left[I + \frac{V_\beta}{100n} \right]^{-1} \sum_{i=1}^n \frac{\beta_i}{n}, \frac{V_\beta}{n} \left[I + \frac{V_\beta}{100n} \right]^{-1} \right)$$

2) Generate V_β

$$[V_\beta | \{\beta_i, b\}] \sim \text{Inverted Wishart} \left[p + n + 2, I + \sum_{i=1}^n (\beta_i - b)(\beta_i - b)^T \right]$$

3) Generate $\{\beta_i, i = 1, \dots, n\}$

$$f(\beta_i | y_i^I, x_i^I, b, V_\beta) \propto \exp \left\{ -\frac{1}{2} (\beta_i - b)^T V_\beta^{-1} (\beta_i - b) \right\} p(y_i^I | x_i^I, \beta_i),$$

$$\text{where } p(y_i^I | x_i^I, \beta_i) = \prod_{q=1}^{12} \frac{\exp \{ \beta_i^T x_{iq, y_{iq}^I}^I \}}{\sum_{k=1}^4 \exp \{ \beta_i^T x_{iq, k}^I \}}.$$

Pseudolikelihood

1) Generate b and V_β the same way as in hierarchical Bayes model.

2) Generate $\{\beta_i, i = 1, \dots, n\}$

i) for $i \leq m$,

$$f(\beta_i | y_i^I, y, x_i^I, x, b, V_\beta, \gamma, g, W) \propto \exp \left\{ -\frac{1}{2} (\beta_i - b)^T V_\beta^{-1} (\beta_i - b) \right\} p(y_i^I | x_i^I, \beta_i) p(y_i | y_{-i}, x, \beta_i, \gamma, g, W)$$

where $p(y_i | y_{-i}, x, \beta_i, \gamma, g, W)$ is the choice probability associated with individual i that depends only on his or her neighbors' choices.

ii) for $i > m$, β_i is generated as in hierarchical Bayes model.

3) Generate $\{\gamma_j, j = 0, \dots, 3\}$

$$f(\gamma_j | y, x, \beta, \gamma_{-j}, g, W) \propto \exp\left\{-\frac{\gamma_j^T \gamma_j}{200}\right\} \prod_{i=1}^m p(y_i | y_{-i}, x, \beta_i, \gamma, g, W)$$

Approximate Sampling

1) Generate b and V_β the same way as in hierarchical Bayes model.

2) Generate $\{\beta_i, i = 1, \dots, n\}$

i) for $i \leq m$,

$$f(\beta_i | y_i^I, y, x_i^I, x, b, V_\beta, \gamma, g, W) \propto \exp\left\{-\frac{1}{2}(\beta_i - b)^T V_\beta^{-1}(\beta_i - b)\right\} p(y_i^I | x_i^I, \beta_i) p(y | x, \beta, \gamma, g, W),$$

where $p(y | x, \beta, \gamma, g, W)$ is defined in (8a and 8b). We need to apply approximate sampling to draw β_i . Please refer to algorithm description in modeling framework and theory section for details.

ii) for $i > m$, β_i is generated as in hierarchical Bayes model.

3) Generate $\{\gamma_j, j = 0, \dots, 3\}$

$$f(\gamma_j | y, x, \beta, \gamma_{-j}, g, W) \propto \exp\left\{-\frac{\gamma_j^T \gamma_j}{200}\right\} p(y | x, \beta, \gamma, g, W),$$

approximate sampling is used to draw γ_j .

Discrete-time Markov Chain

1) Generate b and V_β the same way as in hierarchical Bayes model.

2) Generate $\{\beta_i, i = 1, \dots, n\}$

i) for $i \leq m$,

$$f(\beta_i | y_i^I, x_i^I, s^0, \dots, s^T, b, V_\beta, \gamma, g, W)$$

$$\propto \exp\left\{-\frac{1}{2}(\beta_i - b)^T V_\beta^{-1}(\beta_i - b)\right\} P(y_i^I | x_i^I, \beta_i) \prod_{t: c^{t+1}=i} p(y_i^{t+1} | c^{t+1}=i, c^t, y^t, x, \beta_i, \gamma, g, W),$$

where $p(y_i^{t+1} | c^{t+1}=i, c^t, y^t, x, \beta_i, \gamma, g, W) = \exp\{u_{by_i^{t+1}}^{t+1}\} / \sum_{k=I(y_i^t=0)}^4 \exp\{u_{ik}^{t+1}\}$ with u_{ik}^{t+1} defined in (9a and 9b).

ii) for $i > m$, β_i is generated as in hierarchical Bayes model.

3) Generate $\{\gamma_j, j = 0, \dots, 3\}$

$$f(\gamma_j | s^1, \dots, s^T, x, \beta, \gamma_{-j}, g, W) \propto \exp\left\{-\frac{\gamma_j^T \gamma_j}{200}\right\} \prod_{t=0}^{T-1} p(s^{t+1} | s^t, x, \beta, \gamma, g, W),$$

Where $p(s^{t+1} | s^t, x, \beta, \gamma, g, W) = \frac{1}{m} I(y_{-c^{t+1}}^{t+1} = y_{-c^{t+1}}^t) \exp\{u_{c^{t+1}y_{c^{t+1}}^{t+1}}^{t+1}\} / \sum_{k=I(y_{c^{t+1}}^t=0)}^4 \exp\{u_{c^{t+1}k}^{t+1}\}$ with $u_{c^{t+1}k}^{t+1}$

defined in (9a and 9b).

In-sample Fit Procedures:

We identify the new choices with a superscript *. To make the Markov chain more comparable to other models, we fix who intends to change $\{c^t, t = 1, \dots, T\}$ for in-sample fit, and only generate new choices.

Hierarchical Bayes Model

$$p(y^* | y^I, x^I, \beta) = \prod_{i=1}^n p(y_i^* | y^I, x^I, \beta_i), \text{ where } p(y_i^* | y^I, x^I, \beta_i) = \frac{\exp\{\beta_i^T x_{y_i^*}\}}{\sum_{k=0}^4 \exp\{\beta_i^T x_k\}}.$$

1) Use the posterior draws of $\{\beta_i, i = 1, \dots, m\}$ from the estimation procedure to estimate

$$p(y_i^* | y^I, x^I, \beta_i).$$

2) Generate $\{y_i^*, i = 1, \dots, m\}$ each from $p(y_i^* | y^I, x^I, \beta_i)$.

Pseudolikelihood

As we mentioned earlier, due to interdependence of y^* , we cannot draw y_i^* independently from y_{-i}^* . A more computationally efficient way is to draw y^* within the parameter estimation procedure. Below is the procedure which produces posterior parameter draws along with in-sample predictions.

1) Generate b and V_β the same way as in hierarchical Bayes model.

2) Generate $\{\beta_i, i = 1, \dots, n\}$

i) for $i \leq m$,

$$f(\beta_i | y_i^I, y, y^*, x_i^I, x, b, V_\beta, \gamma, g, W) \\ \propto \exp\left\{-\frac{1}{2}(\beta_i - b)^T V_\beta^{-1}(\beta_i - b)\right\} p(y_i^I | x_i^I, \beta_i) P(y_i | y_{-i}, x, \beta_i, \gamma, g, W) p(y_i^* | y_{-i}^*, x, \beta_i, \gamma, g, W)$$

ii) for $i > m$, β_i is generated as in hierarchical Bayes model.

3) Generate $\{\gamma_j, j = 0, \dots, 3\}$

$$f(\gamma_j | y, y^*, x, \beta, \gamma_{-j}, g, W) \propto \exp\left\{-\frac{\gamma_j^T \gamma_j}{200}\right\} \prod_{i=1}^m p(y_i | y_{-i}, x, \beta_i, \gamma, g, W) \prod_{i=1}^m p(y_i^* | y_{-i}^*, x, \beta_i, \gamma, g, W)$$

4) Generate $\{y_i^*, i = 1, \dots, m\}$

$$p(y_i^* | y_{-i}^*, x, \beta, \gamma, g, W) = \frac{\prod_{j=1}^m p(y_j^* | y_{-j}^*, x, \beta_j, \gamma, g, W)}{\sum_{y_i^*=0}^4 \prod_{j=1}^m p(y_j^* | y_{-j}^*, x, \beta_j, \gamma, g, W)}$$

Approximate Sampling

1) Generate b and V_β the same way as in hierarchical Bayes model.

2) Generate $\{\beta_i, i = 1, \dots, n\}$

i) for $i \leq m$,

$$f(\beta_i | y_i^l, y, y^*, x_i^l, x, b, V_\beta, \beta_{-i}, \gamma, g, W)$$

$$\propto \exp\left\{-\frac{1}{2}(\beta_i - b)^T V_\beta^{-1}(\beta_i - b)\right\} p(y_i^l | x_i^l, \beta_i) p(y | x, \beta, \gamma, g, W) p(y^* | x, \beta, \gamma, g, W)$$

Again, we need to apply approximate sampling to draw β_i .

ii) for $i > m$, β_i is generated as in hierarchical Bayes model.

3) Generate $\{\gamma_j, j = 0, \dots, 3\}$

$$f(\gamma_j | y, x, \beta, \gamma_{-j}, g, W) \propto \exp\left\{-\frac{\gamma_j^T \gamma_j}{200}\right\} p(y | x, \beta, \gamma, g, W) p(y^* | x, \beta, \gamma, g, W),$$

approximate sampling is used to draw γ_j .

4) Generate $\{y_i^*, i = 1, \dots, m\}$

$$p(y_i^* | y_{-i}^*, x, \beta, \gamma, g, W) = \frac{f(y_i^*, y_{-i}^* | x, \beta_i, \gamma, g, W)}{\sum_{v=0}^4 f(v, y_{-i}^* | x, \beta_i, \gamma, g, W)},$$

where $f(y_i^*, y_{-i}^* | x, \beta_i, \gamma, g, W)$ is the un-normalized part of $p(y_i^*, y_{-i}^* | x, \beta_i, \gamma, g, W)$.

Discrete-time Markov Chain

1) Generate b and V_β the same way as in hierarchical Bayes model.

2) Generate $\{\beta_i, i = 1, \dots, n\}$

i) for $i \leq m$,

$$f(\beta_i | y_i^l, x_i^l, s^0, \dots, s^T, y^{1*}, \dots, y^{T*}, b, V_\beta, \gamma, g, W) \propto \exp\left\{-\frac{1}{2}(\beta_i - b)^T V_\beta^{-1}(\beta_i - b)\right\} P(y_i^l | x_i^l, \beta_i) \times$$

$$\prod_{t: c^{t+1}=i} p(y_i^{t+1} | c^{t+1}=i, c^t, y^t, x, \beta_i, \gamma, g, W) \prod_{t: c^{t+1}=i} p(y_i^{(t+1)*} | c^{t+1}=i, c^t, y^{t*}, x, \beta_i, \gamma, g, W)$$

ii) for $i > m$, β_i is generated as in hierarchical Bayes model.

3) Generate $\{\gamma_j, j = 0, \dots, 3\}$

$$f(\gamma_j | s^0, \dots, s^T, y^{1*}, \dots, y^{T*}, x, \beta, \gamma_{-j}, g, W) \\ \propto \exp\left\{-\frac{\gamma_j^T \gamma_j}{200}\right\} \prod_{t=0}^{T-1} p(s^{t+1} | s^t, x, \beta, \gamma, g, W) \prod_{t=0}^{T-1} p(y_{c^{t+1}}^{(t+1)*} | c^{t+1}, c^t, y^{t*}, x, \beta_i, \gamma, g, W).$$

4) Generate $\{y^{(t+1)*}, t = 0, \dots, T-1\}$

$$p(y^{(t+1)*} | c^0, \dots, c^T, y^{1*}, \dots, y^{t*}, y^{(t+2)*}, \dots, y^{T*}, x, \beta, \gamma, g, W) \\ = \frac{\prod_{q=0}^{T-1} p(y_{c^{q+1}}^{(q+1)*} | c^{q+1}, c^q, y^{q*}, x, \beta, \gamma, g, W) I(y_{-c^{q+1}}^{(q+1)*} = y_{-c^{q+1}}^{q*})}{\sum_{y_{c^{t+1}}^{(t+1)*} = I(y_{c^{t+1}}^{t*} > 0)}^4 \prod_{q=0}^{T-1} p(y_{c^{q+1}}^{(q+1)*} | c^{q+1}, c^q, y^{q*}, x, \beta, \gamma, g, W) I(y_{-c^{q+1}}^{(q+1)*} = y_{-c^{q+1}}^{q*})}$$

Out-of-sample Fit Procedures:

For notation simplicity, assume the students are sorted so that the out-of-sample choices belong

to the first k students. Denote the k out-of-sample choices by $y_{miss}^* = (y_{1,miss}^*, \dots, y_{k,miss}^*)$ and the

remaining in-sample choices by $y_{in} = (y_{k+1,in}, \dots, y_{m,in})$. To make the Markov chain more

comparable to other models, we fix who intends to change $\{c^t, t = 1, \dots, T\}$ for out-of-sample fit,

and only generate choices y_{miss}^* . Notice only their raffle choices are taken out of the sample.

Hierarchical Bayes Model

$$p(y_{miss}^* | y^I, x^I, \beta) = \prod_{i=1}^k p(y_{i,miss}^* | y^I, x^I, \beta_i), \text{ where } p(y_{i,miss}^* | y^I, x^I, \beta_i) = \frac{\exp\{\beta_i^T x_{y_{i,miss}^*}^*\}}{\sum_{k=0}^4 \exp\{\beta_i^T x_k\}}.$$

1) Use the posterior draws of $\{\beta_i, i = 1, \dots, k\}$ from the estimation procedure to estimate

$$p(y_{i,miss}^* | y^I, x^I, \beta_i).$$

2) Generate $\{y_{i,miss}^*, i = 1, \dots, k\}$ each from $p(y_{i,miss}^* | y^I, x^I, \beta_i)$.

Pseudolikelihood

As we mentioned earlier, due to interdependence of in-sample and out-of-sample choices, we should treat out-of-sample data as missing data when estimating parameters. A computationally efficient way is to draw y_{miss}^* within the parameter estimation procedure. Below is the procedure which produces posterior parameter draws along with out-of-sample predictions. For Markov chain,

1) Generate b and V_β the same way as in hierarchical Bayes model.

2) Generate $\{\beta_i, i = 1, \dots, n\}$

i) for $i \leq m$,

$$f(\beta_i | y_i^I, y_{in}, y_{miss}^*, x_i^I, x, b, V_\beta, \gamma, g, W) \\ \propto \exp\left\{-\frac{1}{2}(\beta_i - b)^T V_\beta^{-1}(\beta_i - b)\right\} p(y_i^I | x_i^I, \beta_i) p(y_i | y_{-i,in}, y_{-i,miss}^*, x, \beta_i, \gamma, g, W)$$

ii) for $i > m$, β_i is generated as in hierarchical Bayes model.

3) Generate $\{\gamma_j, j = 0, \dots, 3\}$

$$f(\gamma_j | y_{in}, y_{miss}^*, x, \beta, \gamma_{-j}, g, W) \\ \propto \exp\left\{-\frac{\gamma_j^T \gamma_j}{200}\right\} \prod_{i=1}^k p(y_{i,miss}^* | y_{-i,miss}^*, y_{in}, x, \beta_i, \gamma, g, W) \prod_{i=k+1}^m p(y_{i,in} | y_{miss}^*, y_{-i,in}, x, \beta_i, \gamma, g, W)$$

4) Generate $\{y_{i,miss}^*, i = 1, \dots, k\}$

$$p(y_{i,miss}^* | y_{in}, y_{-i,miss}^*, x, \beta, \gamma, g, W) \\ = \frac{\prod_{j=1}^k p(y_{j,miss}^* | y_{-j,miss}^*, y_{in}, x, \beta_j, \gamma, g, W) \prod_{j=k+1}^m p(y_{j,in} | y_{miss}^*, y_{-j,in}, x, \beta_j, \gamma, g, W)}{\sum_{y_{i,miss}^*=0}^4 \prod_{j=1}^k p(y_{j,miss}^* | y_{-j,miss}^*, y_{in}, x, \beta_j, \gamma, g, W) \prod_{j=k+1}^m p(y_{j,in} | y_{miss}^*, y_{-j,in}, x, \beta_j, \gamma, g, W)}$$

Approximate Sampling

1) Generate b and V_β the same way as in hierarchical Bayes model.

2) Generate $\{\beta_i, i = 1, \dots, n\}$

i) for $i \leq m$,

$$f(\beta_i | y_i^I, y_{in}, y_{miss}^*, x_i^I, x, b, V_\beta, \beta_{-i}, \gamma, g, W) \\ \propto \exp\left\{-\frac{1}{2}(\beta_i - b)^T V_\beta^{-1}(\beta_i - b)\right\} p(y_i^I | x_i^I, \beta_i) p(y_{in}, y_{miss}^* | x, \beta, \gamma, g, W)$$

Again, we need to apply approximate sampling to draw β_i .

ii) for $i > m$, β_i is generated as in hierarchical Bayes model.

3) Generate $\{\gamma_j, j = 0, \dots, 3\}$

$$f(\gamma_j | y, x, \beta, \gamma_{-j}, g, W) \propto \exp\left\{-\frac{\gamma_j^T \gamma_j}{200}\right\} p(y_{in}, y_{miss}^* | x, \beta, \gamma, g, W),$$

approximate sampling is used to draw γ_j .

4) Generate $\{y_{i,miss}^*, i = 1, \dots, k\}$

$$p(y_{i,miss}^* | y_{in}, y_{-i,miss}^*, x, \beta, \gamma, g, W) = \frac{f(y_{in}, y_{i,miss}^*, y_{-i,miss}^* | x, \beta_i, \gamma, g, W)}{\sum_{v=0}^4 f(y_{in}, v, y_{-i,miss}^* | x, \beta_i, \gamma, g, W)},$$

where $f(y_{in}, y_{i,miss}^*, y_{-i,miss}^* | x, \beta_i, \gamma, g, W)$ is the un-normalized part of

$$p(y_{in}, y_{i,miss}^*, y_{-i,miss}^* | x, \beta_i, \gamma, g, W).$$

Discrete-time Markov Chain

Denote the choices at the k missing time points by $y^{miss,*} = (y^{miss,t_1^*}, \dots, y^{miss,t_k^*})$. They are not

necessarily the first k time points. The in-sample choices are y^{in} .

1) Generate b and V_β the same way as in hierarchical Bayes model.

2) Generate $\{\beta_i, i = 1, \dots, n\}$

i) for $i \leq m$,

$$f(\beta_i | y_i^I, x_i^I, c^0, \dots, c^T, y^{in}, y^{miss*}, b, V_\beta, \gamma, g, W) \propto \exp\left\{-\frac{1}{2}(\beta_i - b)^T V_\beta^{-1}(\beta_i - b)\right\} P(y_i^I | x_i^I, \beta_i) \times$$

$$\prod_{t: c^{t+1}=i} p(y_i^{t+1} | c^{t+1}=i, c^t, y^t, x, \beta_i, \gamma, g, W),$$

note that y^t and y_i^{t+1} could be out-of-sample choices, for notation lucidity we do not specify it here.

ii) for $i > m$, β_i is generated as in hierarchical Bayes model.

3) Generate $\{\gamma_j, j = 0, \dots, 3\}$

$$f(\gamma_j | c^0, \dots, c^T, y^{in}, y^{miss*}, x, \beta, \gamma_{-j}, g, W) \propto \exp\left\{-\frac{\gamma_j^T \gamma_j}{200}\right\} \prod_{t=0}^{T-1} p(s^{t+1} | s^t, x, \beta, \gamma, g, W),$$

again we do not explicitly specify out-of-sample choices here.

4) Generate $\{y^{miss, t_i^*}, i = 1, \dots, k\}$

$$p(y^{miss, t_i^*} | c^0, \dots, c^T, y^{in}, y^{miss, -t_i^*}, x, \beta, \gamma, g, W)$$

$$= \frac{\prod_{q=0}^{T-1} p(y_{c^{q+1}}^{(q+1)} | c^{q+1}, c^q, y^q, x, \beta, \gamma, g, W) I(y_{-c^{q+1}}^{(q+1)} = y_{-c^{q+1}}^q)}{\sum_{y_{c^{t+1}}^{miss, t_i^*} = I(y_{c^{t+1}}^{t_i-1} > 0)}^4 \prod_{q=0}^{T-1} p(y_{c^{q+1}}^{(q+1)} | c^{q+1}, c^q, y^q, x, \beta, \gamma, g, W) I(y_{-c^{q+1}}^{(q+1)} = y_{-c^{q+1}}^q)},$$

again y^q could be out-of-sample choices.

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